ZEWAIL CITY OF SCIENCE, TECHNOLOGY AND INNOVATION

School of Computational Sciences and Artificial Intelligence

CSAI 203: Introduction to Software Engineering

INTERNSHIP & CAREER PORTAL

Phase 2: Software Requirements Specification (SRS)

A Web-Based Platform Connecting Academic Talent with Corporate Recruitment

Course Instructor:	Dr. Mohamed Sami Rakha
Academic Term:	Fall 2026
Team Identifier:	Team 05
Representative ID:	202507986
Representative Email:	s-nour.ahmed@zewailcity.edu.eg
Team Members:	Hagar Mohamed Ahmed (ID: 202508608) Salma Ashraf Ibrahim (ID: 202500842) Nour Ahmed Mohamed (ID: 202507986)
Submission Date:	October 24, 2026

TABLE OF CONTENTS

Section	Page
1. Introduction (Purpose, Scope, Definitions, References, Overview)	3
2. Overall Description (Perspective, Functions, Users, Stakeholders, Environment, Constraints)	4
3. Specific Requirements	6
3.1 Functional Requirements (FR-01 to FR-20 Detailed Scenarios)	6
3.2 Use Case Model (3 Subsystem Diagrams & 10 IEEE Descriptions)	8
3.3 Domain Model (Conceptual Class Diagram & Descriptions)	11
3.4 Non-Functional Requirements & Testing Criteria	12
3.5 External Interface Requirements	13
4. Appendices (Appendix A: Data Dictionary, Appendix B: Glossary)	14

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the Internship & Career Portal. It details all functional capabilities, operational rules, system constraints, user interactions, and quality attributes governing the software system. Intended for project stakeholders, software developers, quality assurance engineers, and academic evaluators, this specification serves as the formal baseline for system architecture, testing, and implementation during the Fall 2026 academic term.

1.2 Scope

The Internship & Career Portal is a centralized web platform designed to connect undergraduate students seeking industry opportunities with corporate employers searching for talent. The platform standardizes the recruitment pipeline by replacing fragmented, manual channels with an automated end-to-end management workflow.

Core Capabilities: The platform enables students to build academic profiles, search verified internship listings, submit applications with stored CV document links, and track real-time application statuses. Employers receive structured recruitment tools to post opportunities, review applicant portfolios, update selection statuses, and schedule structured interviews. System Administrators maintain platform integrity through content moderation and account management.

Out of Scope: Direct financial payment processing for premium listings, built-in video conferencing streaming tools, and automated background checks are explicitly excluded from the current system release.

1.3 Definitions, Acronyms, and Abbreviations

A complete list of domain-specific terms, technical acronyms, and software engineering abbreviations is provided in Appendix B (Glossary) on page 14 of this document.

1.4 References

1. CSAI 203: Introduction to Software Engineering Project Guidelines, Zewail City of Science, Technology and Innovation, Fall 2026.
2. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, IEEE Computer Society, 1998.

1.5 Overview

The remainder of this document is organized into three major sections. Section 2 describes high-level system context, user roles, operational environment, and implementation constraints. Section 3 details functional requirements with thorough scenarios, UML Use Case models, IEEE Use Case tables, Conceptual Class Diagram, non-functional quality goals, and external interfaces. Section 4 contains the Data Dictionary and Glossary.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Internship & Career Portal is an independent, self-contained web application operating under a multi-tier Model-View-Controller (MVC) architecture. The backend service processes incoming HTTP requests, enforces business domain rules, manages authentication sessions, and interacts with a relational database engine. The presentation layer serves responsive HTML5/CSS web interfaces rendered directly within client web browsers.

2.2 Product Functions

The system executes six core functional workflows:

- **User Management & Authentication:** Role-based account creation, credential verification, secure session management, and password recovery via tokenized links.
- **Student Profile & Career Search:** Profile management, searchable internship directories, and multi-criteria filtering by industry, employment type, and location.
- **Application Pipeline:** Eligibility verification, standardized application submission with CV link attachment, real-time tracking, and formal application withdrawal.
- **Employer Listing Management:** Publishing, editing, closing, and archiving internship postings with configurable deadlines.
- **Candidate Evaluation & Interview Scheduling:** Application reviewing, candidate status updates (Shortlist, Accept, Reject), and interview session scheduling with meeting link generation.
- **System Moderation:** Administrative oversight for job listing validation, user account management (suspend/reactivate), and policy enforcement.

2.3 User Classes and Characteristics

- **Students:** Primary end-users who access the portal via mobile or desktop web browsers to search for opportunities, manage academic profiles, apply for positions, and monitor selection progress.
- **Employers (Corporate Recruiters):** Business end-users requiring intuitive interfaces to publish job requirements, evaluate applicant profiles, communicate hiring decisions, and organize interview timelines.
- **System Administrators:** Operations operators responsible for platform safety with elevated privileges to audit platform activity, delete policy-violating listings, and manage user accounts.

2.4 Software Stakeholders

Key stakeholders include undergraduate students seeking professional experience, corporate hiring partners recruiting talent, course instructors and TAs evaluating software engineering practices, and the development team (Team 05).

2.5 Operating Environment

The system operates across cross-platform configurations: Client devices (PCs, laptops, tablets, smartphones) with internet connection and HTML5 web browsers (Google Chrome v100+, Mozilla Firefox v98+, Apple Safari v15+). The server runs on a Linux/Unix or Windows environment executing an instructor-approved backend runtime paired with a relational database engine (MySQL or SQLite).

2.6 Design and Implementation Constraints

- **Architectural Pattern:** Mandatory implementation of the Model-View-Controller (MVC) pattern to separate domain logic, UI rendering, and request routing.
- **Technology Scope:** Restricted to instructor-approved backend frameworks, relational databases, and standard web technologies (HTML5, CSS3, vanilla JS). Heavy front-end frameworks are avoided.
- **Resource Constraints:** Developed and documented by a team of 3 engineering students within an 8-week implementation window.

2.7 User Documentation

The portal includes an integrated Frequently Asked Questions (FAQ) page accessible directly from the main navigation header, guiding users through account setup, CV formatting, application tracking, and posting rules.

2.8 Assumptions and Dependencies

Assumptions: Users have continuous internet connectivity; student applicants possess accessible, externally hosted CV document URLs (e.g., Google Drive or LinkedIn link) prior to applying.

Dependencies: Availability and stability of HTTPS protocols; continuous uptime of the backend relational database service.

3. SPECIFIC REQUIREMENTS

3.1 Functional Requirements

FR-01: Student Account Registration

The system shall allow new students to register by providing their full name, academic email (.edu domain), password, university, major, and expected graduation year. *Scenario:* Upon submission, the system validates email uniqueness and password complexity. If valid, an account is created; if the email exists, an error message is displayed.

FR-02: User Authentication (Login)

The system shall authenticate students, employers, and admins using email and password. *Scenario:* Correct credentials grant access to the role-specific dashboard. Three failed attempts trigger a 5-minute lockout notice.

FR-03: Password Reset Request

The system shall provide a password recovery mechanism via email reset tokens. *Scenario:* The user enters their registered email. The system sends a single-use token valid for 15 minutes to reset the password.

FR-04: Edit Student Profile

FR-06: Search Internship Postings

The system shall allow users to perform free-text searches against job titles, company names, and descriptions. *Scenario:* Entering "Software" retrieves all active postings containing the term. If no matches are found, a "No results found" guidance prompt is displayed.

FR-07: Filter Internship Postings

The system shall filter listings based on criteria: location (On-site, Hybrid, Remote), industry, and duration. *Scenario:* A student selects "Remote" and "Engineering"; the list updates dynamically to reflect matching postings.

FR-08: View Internship Detailed Information

The system shall display full details for a selected internship, including full description, requirements, stipend, location, and recruiter contact info. *Scenario:* Clicking a listing opens the details page with an active "Apply Now" button.

FR-09: Apply for Internship Position

The system shall allow eligible students to apply for an open internship by providing an optional cover letter alongside their stored profile CV link. *Scenario:* The student confirms submission; the system verifies eligibility, records the application status as "Pending", and notifies the student.

FR-10: Track Submitted Applications

The system shall provide a personal student dashboard displaying all submitted applications and their live status (Pending, Shortlisted, Accepted, Rejected). *Scenario:* The student logs in to review real-time feedback from employers.

FR-11: Withdraw Submitted Application

The system shall permit students to withdraw a pending application prior to employer review. *Scenario:* Clicking "Withdraw" updates application status to "Withdrawn" and removes the candidate from the employer's active review queue.

FR-12: Edit Company Profile

The system shall allow employers to manage corporate details, including company name, logo URL, industry sector, website, and company description. *Scenario:* Updates immediately reflect on all associated job postings.

FR-13: Post New Internship Opportunity

The system shall enable verified employers to publish new listings specifying title, requirements, location, work type, stipend, and deadline. *Scenario:* Valid entries make the posting immediately visible in the active student catalog.

FR-14: Edit Existing Internship Posting

The system shall allow employers to modify active listings under their company account. *Scenario:* An employer extends an application deadline; the system updates the database and displays the updated date to browsing students.

FR-15: Close or Delete Internship Posting

The system shall allow employers to close applications or delete a listing. *Scenario:* Closing a listing sets status to "Closed", preventing new applications while preserving historical records for existing applicants.

FR-16: View Job Applicants List

The system shall allow employers to view all candidate applications for their postings, including student profiles, cover letters, and CV links. *Scenario:* The employer filters candidates by application date or status.

FR-17: Accept or Reject Job Applicants

The system shall enable employers to update applicant status to "Shortlisted", "Accepted", or "Rejected". *Scenario:* Changing status updates the student's application tracking dashboard instantly.

FR-18: Schedule Candidate Interview

The system shall allow employers to schedule interviews with shortlisted applicants by defining date, time, and meeting details. *Scenario:* Submitting the form creates an interview record linked to the application and sends meeting details to the candidate dashboard.

FR-19: Moderate Internship Postings

The system shall empower administrators to review, flag, or remove listings violating guidelines. *Scenario:* An admin flags a fraudulent job posting, instantly hiding it from student search results and notifying the employer.

FR-20: Manage User Accounts

The system shall allow administrators to view all accounts, suspend abusive users, or reactivate profiles. *Scenario:* An admin suspends an abusive account, immediately revoking login authorization across all sessions.

3.2 Use Case Model

The system functionality is structured into three subsystem UML Use Case diagrams representing key operational areas.

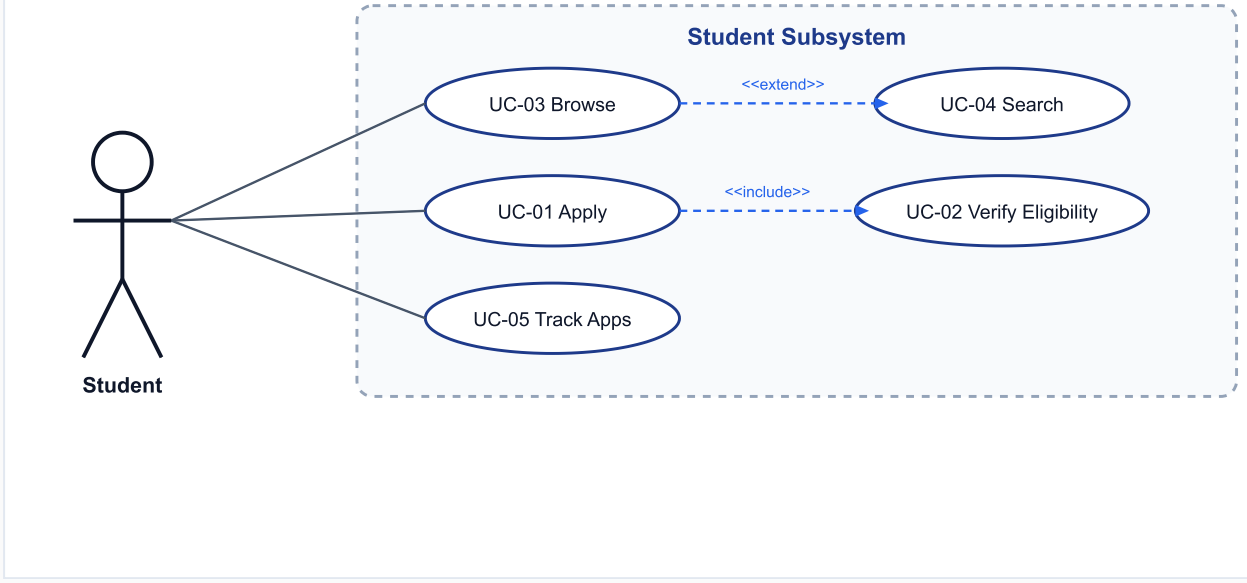


Figure 1: Student Subsystem Use Case Diagram

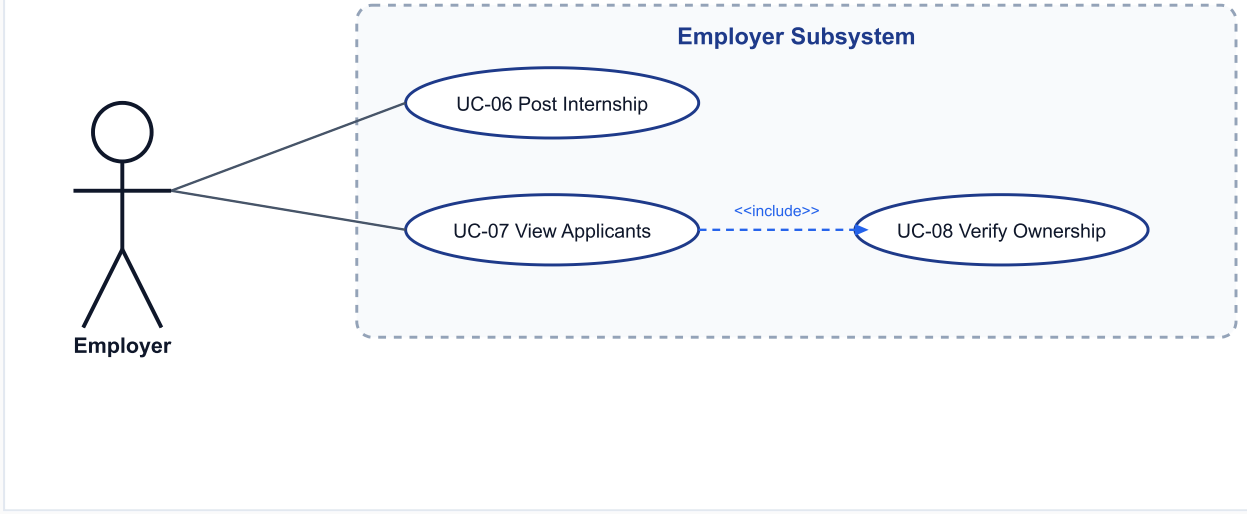


Figure 2: Employer Subsystem Use Case Diagram

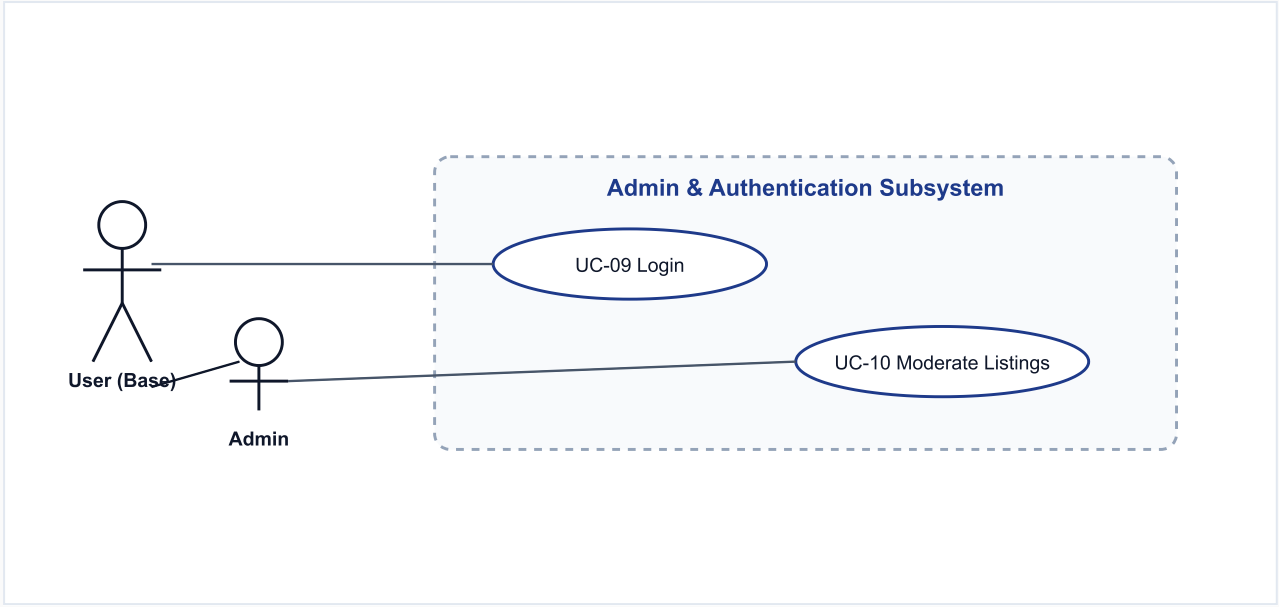


Figure 3: Admin & Authentication Subsystem Use Case Diagram

3.2.2 Complete Use Case IEEE Descriptions

Use Case ID/Name	**UC-01: Apply for Internship**
Primary Actor	Student
Preconditions	Student is authenticated and viewing an active, open internship posting page.
Postconditions	Application record is created with status "Pending" and stored in the database.
Main Success Scenario	1. Student clicks "Apply Now". 2. System executes <<include>> UC-02 Verify Eligibility. 3. System displays application modal with stored profile CV link. 4. Student enters optional cover letter and clicks "Submit". 5. System saves application and displays success confirmation.
Extensions	2a. Student has missing CV or prior active application: System displays eligibility error alert and aborts workflow.

Use Case ID/Name	**UC-02: Verify Eligibility (Included Use Case)**
Primary Actor	Automated System Subsystem
Preconditions	Triggered automatically during UC-01 application process.
Postconditions	System validates or denies student authorization to apply.
Main Success Scenario	1. System verifies student profile contains a non-empty, valid CV link. 2. System checks internship application deadline hasn't passed. 3. System checks database for active prior applications by this student for this position. 4. System returns positive confirmation to UC-01.

Use Case ID/Name	**UC-03: Browse Internships**
Primary Actor	Student / Guest User
Preconditions	Portal application is accessible online.
Postconditions	Catalog of open internship positions is displayed.
Main Success Scenario	1. User navigates to Internships page. 2. System fetches all active, non-expired listings from database.

Use Case ID/Name	**UC-04: Search & Filter Internships (Extended Use Case)**
Primary Actor	Student / Guest User
Preconditions	User is currently executing UC-03 Browse Internships.
Postconditions	Filtered subset of matching internship listings is rendered.
Main Success Scenario	1. User enters free-text query or selects filter checkboxes (Location, Industry, Work Type). 2. System queries database against title, company, and criteria tags. 3. System updates catalog view with matching active records.
Extensions	2a. No listings match criteria: System renders "No internships found matching your criteria" message with a filter reset button.

Use Case ID/Name	**UC-05: Track Applications**
Primary Actor	Student
Preconditions	Student is logged in.
Postconditions	List of all submitted applications with live statuses is displayed.
Main Success Scenario	1. Student accesses "My Applications" dashboard. 2. System retrieves all application records linked to student ID. 3. System displays table showing company, position, date submitted, and status (Pending, Shortlisted, Accepted, Rejected).

Use Case ID/Name	**UC-06: Post Internship**
Primary Actor	Employer
Preconditions	Employer is logged in with verified account status.
Postconditions	New internship listing is stored and published to active catalog.
Main Success Scenario	1. Employer clicks "Post New Internship". 2. System displays job creation form. 3. Employer inputs job title, description, requirements, location, work type, stipend, and deadline. 4. Employer submits form. 5. System validates required fields and saves new listing.
Extensions	4a. Required fields missing or invalid deadline date: System highlights errors and requests re-entry.

Use Case ID/Name	**UC-07: View Applicants**
Primary Actor	Employer
Preconditions	Employer is logged in and has created internship postings.
Postconditions	List of candidate applications for selected position is rendered.
Main Success Scenario	1. Employer selects job posting from dashboard. 2. System executes <<include>> UC-08 Verify Ownership. 3. System retrieves and displays list of applicant names, submission dates, cover letters, and CV links.

Use Case ID/Name	**UC-08: Verify Ownership (Included Use Case)**
Primary Actor	Automated System Subsystem
Preconditions	Triggered automatically when employer attempts to view applicants (UC-07) or edit listings (FR-14).
Postconditions	System authorizes or denies employer access to target listing data.
Main Success Scenario	1. System retrieves employer ID from active session. 2. System compares session employer ID against foreign key stored on target internship listing. 3. System confirms ownership match and returns authorization token.
Extensions	3a. Employer ID does not match listing owner: System logs security violation and displays "403 Unauthorized Access" page.

Use Case ID/Name	**UC-09: User Login**
Primary Actor	User (Student, Employer, Admin)
Preconditions	User has a registered account in the system.
Postconditions	User is authenticated and session token is assigned based on role.
Main Success Scenario	1. User enters registered email and password on login page. 2. System retrieves user record by email and verifies hashed password. 3. System identifies user role and redirects to corresponding dashboard.
Extensions	2a. Incorrect password or unverified email: System displays "Invalid credentials" notice.

Use Case ID/Name	**UC-10: Moderate Listings**
Primary Actor	System Administrator
Preconditions	Admin is logged in with administrative privileges.
Postconditions	Target posting is flagged, edited, or permanently deleted.
Main Success Scenario	1. Admin reviews flagged listings queue in Admin Console. 2. Admin selects listing violating platform terms. 3. Admin selects "Flag" or "Remove Listing". 4. System updates listing status to "Suspended" and notifies posting employer via email.

3.3 Domain Model

The Domain Model captures real-world entities, attributes (excluding technical data types per IEEE standards), generalization hierarchies, and multiplicities governing domain relationships.

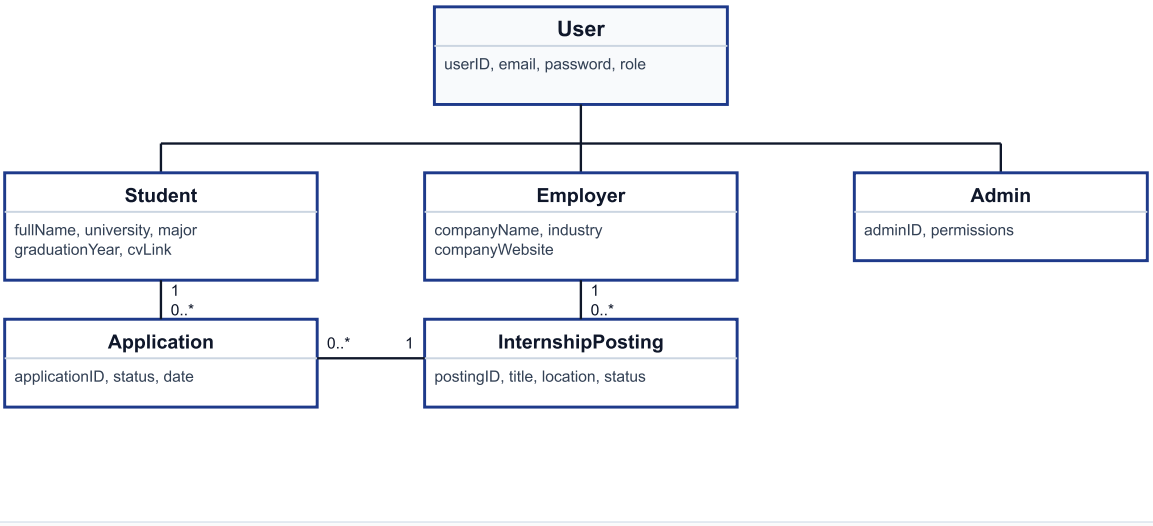


Figure 4: Conceptual Domain Class Diagram

3.3.2 Domain Class Descriptions

1. **User:** Abstract parent domain entity encapsulating system-wide credentials (userID, email, password, role).
2. **Student:** Subclass representing student job seekers. Attributes: fullName, university, major, graduationYear, cvLink. Associations: Holds a 1-to-many relationship with Application (a student can submit 0 or more applications).
3. **Employer:** Subclass representing corporate recruiters. Attributes: companyName, industry, companyWebsite. Associations: Holds a 1-to-many relationship with InternshipPosting (creates and manages multiple listings).
4. **Admin:** Subclass representing platform moderators. Attributes: adminID, permissions. Purpose: Manages user accounts and moderates job content.
5. **InternshipPosting:** Entity representing published positions. Attributes: postingID, title, location, status. Associations: Belongs to 1 Employer; receives 0 or more Applications.
6. **Application:** Linking entity between Student and InternshipPosting. Attributes: applicationID, status, date. Associations: Connects exactly 1 Student to 1 InternshipPosting.

3.4 NON-FUNCTIONAL REQUIREMENTS

System quality attributes are defined through five quantifiable Non-Functional Requirements (NFRs), complete with testing plans and measurable success criteria for verification.

ID / Quality	Requirement Specification	Testing Plan & Success Criteria
NFR-01 Performance	The system backend shall process and render web page responses within 2.0 seconds for 95% of standard HTTP requests under a simulated load of 100 concurrent active user sessions.	**Testing Plan:** Automated load testing using Apache JMeter generating concurrent GET/POST requests. **Success Criteria:** 95th percentile response time ≤ 2.0s with 0 dropped HTTP connections.
NFR-02 Security	All user passwords shall be securely hashed using BCrypt with an appropriate work factor (salt factor 10) prior to database persistence. Role-Based Access Control (RBAC) must strictly enforce endpoint access rules.	**Testing Plan:** Security code audit and penetration test attempting unauthorized URL endpoint access without valid role session cookies. **Success Criteria:** 100% of passwords stored as BCrypt hashes; 0 unauthorized access breaches permitted.
NFR-03 Usability	The portal web interface shall render fluidly across viewport widths ranging from 320px (mobile) to 1920px (desktop). Novice users shall complete core tasks without training.	**Testing Plan:** Usability Acceptance Testing (UAT) with 10 student test participants attempting application submission on mobile devices. **Success Criteria:** ≥ 80% first-attempt task completion rate without intervention.
NFR-04 Reliability	The system shall gracefully catch runtime exceptions and invalid user inputs without displaying raw stack traces or HTTP 500 error pages to end-users.	**Testing Plan:** Negative integration testing submitting malformed HTTP payloads and SQL injection strings. **Success Criteria:** 100% of handled errors display user-friendly notice pages; zero raw stack traces exposed.
NFR-05 Data Integrity	The relational database shall strictly enforce Foreign Key constraints across all table relationships, preventing orphan records upon entity updates or deletions.	**Testing Plan:** Automated SQL integration suite executing cascade deletion tests on employer records linked to active job postings. **Success Criteria:** Referential integrity maintained 100% without orphan application records.

3.5 EXTERNAL INTERFACE REQUIREMENTS

3.5.1 User Interfaces

The portal provides a responsive, accessible graphical user interface (GUI) designed in compliance with standard web usability guidelines. The UI layout consists of four primary global structures:

- **Navigation Header:** Displays permanent platform branding, role-specific navigation menu links (e.g., Browse Internships, My Applications, Post Internship), and authentication status controls (Login, Register, Logout).
- **Student Dashboard:** A structured workspace presenting student profile summaries, saved CV links, and a dynamic data table tracking submitted applications and interview notices.
- **Employer Portal:** An executive recruiter management view featuring job posting creation forms, listing status toggles (Active/Closed), and candidate review queues.
- **Admin Management Console:** A control panel displaying system moderation queues, flagged job posts, and user account suspension tools.

3.5.2 Hardware Interfaces

The system requires no specialized physical hardware devices. It operates over standard internet hardware infrastructure, interfacing with client device input/output peripherals (keyboards, touchscreens, desktop monitors) via HTML5 web browser rendering engines.

3.5.3 Software Interfaces

The application interfaces with two key external software environments:

- **Relational Database Management System (RDBMS):** The MVC controller layer interfaces with an SQL database (e.g., MySQL or SQLite) via native drivers to persist user profiles, credentials, listings, applications, and interview schedules.
- **Client Web Browsers:** The front-end view layer interfaces with client OS environments (Windows, macOS, Linux, Android, iOS) through standard HTML5, CSS3, and JavaScript rendering engines provided by modern browsers.

3.5.4 Communication Interfaces

All client-server network interactions occur over standard Internet Protocol (IP) networks utilizing the HTTPS (HTTP Secure) protocol operating over TLS on TCP port 443. Client-server data exchange utilizes standard HTTP request verbs (GET, POST, PUT, DELETE) with JSON or URL-encoded payloads.

4. APPENDICES

4.1 Appendix A: Data Dictionary

This section defines the relational database schema structure supporting the system features.

Table Name	Key Constraints	Description & Key Field Attributes
users	PK: userID	Stores base account credentials (userID, email, password_hash, role, created_at).
students	PK: studentID FK: userID	Stores student profiles (studentID, userID, full_name, university, major, graduation_year, cv_link).
employers	PK: employerID FK: userID	Stores company recruiter profiles (employerID, userID, company_name, industry, website_url, description).
internships	PK: postingID FK: employerID	Stores internship job listings (postingID, employerID, title, description, location, work_type, stipend, deadline, status).
applications	PK: applicationID FK: postingID, studentID	Links student applications to job postings (applicationID, postingID, studentID, cover_letter, status, applied_at).
interviews	PK: interviewID FK: applicationID	Stores interview schedules (interviewID, applicationID, interview_date, interview_time, location_details, status).

4.2 Appendix B: Glossary

- **Actor:** A person, organization, or automated system that interacts directly with the software system to fulfill a specific goal.
- **BCrypt:** A salted password-hashing function designed to safeguard user credentials against brute-force attacks.
- **MVC Pattern:** Model-View-Controller; an architectural pattern separating business logic (Model), UI presentation (View), and HTTP request routing (Controller).
- **RBAC:** Role-Based Access Control; a security mechanism restricting endpoint privileges based on user roles (Student, Employer, Admin).
- **SRS:** Software Requirements Specification; a formal engineering document detailing system capabilities, rules, and quality constraints.
- **Use Case:** A sequence of actions performed by a system yielding an observable result of value to an actor.